

Perceptual Modifications in Augmented Reality: A Short Survey

Anton Nijholt

Human Media Interaction, University of Twente, The Netherlands
a.nijholt@utwente.nl

ABSTRACT

Views on Augmented Reality (AR) and its technology have changed. Rather than being a research area on its own, with a focus on computer vision and computer-generated imagery, we now have to deal with the integration of AR technology with ubiquitous computing where we have sensors, actuators, and processing capabilities beyond what is possible with a single AR device. The plethora of AR applications that follow from this point of view has not yet fully been investigated. In this paper, we review how in reported AR research this issue is dealt with. We survey the attempts from the AR community to integrate perception technology with AR. These attempts should be added to and integrated with augmented human and multisensorial research that nowadays is being done in the context of ubiquitous computing (Internet of Things, Ambient Intelligence, smart environments).

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality.

KEYWORDS

Augmented reality, augmented humans, perceptual modifications, diminished reality, ever-present augmented reality, eyewear, ubiquitous computing, smart environments, societal concerns

ACM Reference Format:

Anton Nijholt. 2022. Perceptual Modifications in Augmented Reality: A Short Survey. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3544793.3561318>

1 INTRODUCTION

In Augmented Reality (AR) virtual objects are added to the physical environment that we perceive through our AR eyewear or handheld. The virtual objects are not necessarily perceived as virtual. They can be generated and aligned with the environment in a perfect way, indistinguishable from real-world objects. We can also add “virtual skills” to our perceptual skills to enhance them such as being able to see through walls, see behind us, magnify and zoom in on objects, see our environment from above, or see sounds beyond the frequency of the human ear visualized. Hence, there is an artificial modification of a human’s perceptual skills and the perception

of the physical environment, whether or not inhabited by virtual objects.

As usual, we follow Ronald T. Azuma’s AR definition [1], that is, AR combines real and virtual objects in a real environment (1), registers (aligns) real and virtual objects (2), and runs interactively, in three dimensions, and in real-time (3). Moreover, in 2019 Azuma [2] predicted that AR technology, in particular with optical see-through glasses, will be the dominant platform and interface, supplanting the smartphone, for accessing digital information. Smart glasses are seen as “... the best chance of achieving the long-term vision of ubiquitous consumer AR displays.” Although AR is usually vision-oriented, this definition is meant to apply to all senses that can be addressed by content in the augmentation layer that is superimposed on a real, physical, environment. The content can add or support visual, auditory, tactile, olfactory, and gustatory experiences or combinations of these experiences to the existing world.

Our AR technology and devices allow us to perceive and experience this content brought to us integrated with the world that we can perceive without this technology. However, existing, non-AR technology, already provides us with ways to extend our perception and interaction skills and use them to extend or diminish our reality. In this paper, we illustrate through examples how AR technology has made use of other perception technologies to provide an extended augmented reality world, where the enhanced skills to perceive the real world are integrated with AR technology.

In sections 2 and section 3 we discuss AR research in which the use of AR see-through eyewear is integrated with technology that extends the skills with which we perceive the real world. In section 2 the enhancements aim at perceiving more of the world than is possible with the regular use of our senses. In section 3 the enhancements make it possible to leave out or replace the parts of the world that we are not interested in or that interfere with our daily activities. Some societal concerns are mentioned in section 4, followed by a short section with conclusions.

2 PERCEPTUAL MODIFICATION: EXTENDING REALITY

We can extend the capabilities of the cameras that are embedded in AR eyewear. Various magnification ideas for low-vision users have been investigated for OST AR [3]. An AR headset can have both rear and front cameras or a camera that can provide the AR display with a 360 degrees view of our environment. Relevant information from the environment, perhaps collected from sensors in the environment can be visualized on an AR display. Visualization techniques for both in-view and out-of-view nearby objects are surveyed in [4]. An infrared thermography camera is used in [5] to align virtual information and temperature on objects in the environment. Remote cameras, including camera drones, can feed OST eye-wear with visual information from above and from far away. We can have a temporary shift from an egocentric view to



This work is licensed under a Creative Commons Attribution International 4.0 License.

UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom
© 2022 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9423-9/22/09.
<https://doi.org/10.1145/3544793.3561318>

an exocentric bird's-eye view that helps users to determine their location and become more aware of what is happening in the remote environment that they perceive, despite it can not be perceived without the (extended) AR device's capabilities.

Interaction with occluded objects is discussed in [6]. A remote camera view of the object can be presented in the user's OST display at its actual position. The user sees through the occluder (a see-through view) and can physically manipulate the touchable object while it remains out of view. This simulated X-ray vision can also be done with cameras at a larger distance. When our car has access to sensors (cameras) in the urban environment it can receive visual information in an OST display on its windshield about not yet visible vehicles approaching an intersection [7]. The vehicles are still invisible, hidden behind buildings, but the driver can anticipate and respond to possible conflicts. In future smart urban environments, we can expect cameras to provide such information to car drivers or self-driving cars. Remote cameras can track a full scene and have it visualized on the OST AR display. For example, in [8] a camera drone is used to provide a view of an occluded indoor environment and its video frames are texture-mapped on a three-dimensional reconstruction of the indoor environment. In this way, the rebuilt environment allows the operator a 3D immersive view. Similar ideas, also with the HoloLens device, can be found in [9].

To visually slow-motion our view of a changing environment is the topic of research in [10]. Here the AR device (Microsoft's HoloLens) enables user-triggered slow-motion, displaying a decelerated real-time video capture, altering the speed of reality. How to catch up with the continuing real-world happenings and how user interaction in the slow-motion view can be synched with real-time events are among their research questions. Less extreme than in this case, but questions about maintaining context also need to be answered for other interactions that use amplified perceptions, such as the transitions that are required from an egocentric view to an exocentric view of the environment.

Other senses (e.g., auditory, tactile) than the sight sense can be enhanced as well. That is, sound frequencies beyond the human ear can be visualized, and smell sensors can enhance the human smell sense or show the enhancement in a different modality, et cetera. Enhancing one of our senses may require the enhancement of other senses as well to be believable when believability is an issue. In an entertainment, educational, or professional environment believability can be traded for fun and humor, educational aims, and efficiency.

3 PERCEPTUAL MODIFICATION: DIMINISHING REALITY

Diminished Reality (DR) technology is used to remove or conceal real-life sensory experiences from a user's environment. We can look at the technology that reduces or cheats our sensorial capabilities. With oven gloves, we can handle temperatures that we normally can not. Our sunglasses reduce glare. A hearing AR device that reduces background noise and filters out conversations enhances and reduces our sensorial capabilities. Hence, diminishing reality may help us to only pay attention to things that are relevant to us, leaving out all kinds of sensorial clutter. In DR we are not able to perceive reality as it is without augmented reality devices.

In [11] four DR techniques are mentioned: "diminish", "see-through", "replace", and "inpaint". Degrading or distorting the real-world scene is seen as "diminishing", making objects invisible by replacing them with their real-world background is considered to be "see-through", making an object invisible by overlapping it with a virtual object of the same or larger size is the function of the "replace" technique, and "inpaint" refers to a technique where the real object is replaced by a plausible background based on its visible surroundings. Changing the colors of real-world objects is an example of diminishing. To diminish the distinction between real and virtual objects in an AR view, rather than try to make virtual objects indistinguishable from real objects, we can as well try to make the real-world objects in our AR view less perfect and make them look less perfect and more computer-generated [12].

We cannot remove a real object from reality, but we can make it invisible to the user by covering it with virtual objects that represent the real object's background as it can be seen from different users' positions. Usually, generating believable background images requires multiple cameras observing the background, synthesizing the images while knowing about the tracked user's pose (position and orientation), and aligning the background's images in real-time with the user's AR view. If there are limitations to a user's pose changes and views on the background, background images can be prepared in advance and overlaid on the image that needs to be masked or made invisible. Hence, a pre-captured environment model of the real environment is available, and changes in the physical environment are limited, can be controlled, and can be predicted. In this way, we can render environments at recorded poses and replace, for example, a real object with a see-through wireframe-based visualization that replaces the occluding object. We see through the occluder, and at the same time see the occluder's appearance and position at our AR display.

Real-time background generation using multiple cameras or "inpaint" requires image processing that nowadays can only be realized in VST AR where we have more control over the mixing and display of virtual and real content. As mentioned previously, the rendered virtual content on an OST display is not opaque, meaning that the real object remains visible, especially in an outdoor setting. Therefore, most DR applications are built with VST displays. Moreover, in ever-present AR we cannot expect that there will be cameras accompanying us during our outdoor activities and have them record backgrounds of (dynamic) real objects to allow us to have a DR view of our physical environment. However, having AR technology embedded in smart environments makes it possible to use embedded sensors such as cameras to render background images that can be used to replace real content. Rather than performing image editing (e.g. DR inpainting or blurring of sensitive information), recent research in camera technology and deep learning shows the ability to define classes of objects that are optically imaged with high fidelity while erasing other types of objects [13]. Potential applications for AR have yet to be explored.

Finally, the "replace" technique makes it possible to cover or replace unwanted real objects. In OST and VST AR we can control virtual content. It has been argued that in VST AR not only virtual content but also real content can be manipulated. In [14] the possibility is considered to "cut" out real 3D objects and move them as virtual objects through the AR environment. This requires a model

of the environment and real-time adjustments to present the user with an AR view in which these virtual real-world changes take effect without introducing incorrect occlusions.

There are more “down-to-earth” examples of DR realized with OST DR. For example, in remote conversations face detection makes it possible to cover a face with a face-sized emotion-showing emoji to reduce appearance bias [15]. In a multi-user game environment face detection makes it possible to replace a human participant’s face with a mask displaying the face of their favorite game character. In Dr. Grordbort’s Invaders game [16] Magic Leaps eye-tracking technology is used to mirror the player’s eye movements on the mask.

4 PERCEPTUAL MODIFICATIONS: SOCIETAL CONCERNS

In [17], it is mentioned that AR eyeglasses can be designed to filter out and cover or change billboard advertisements or get rid of other visual clutter in urban environments. Similar “replace” ideas can be found in the “White Christmas” episode of the sci-fi Black Mirror series where people can be “blocked”, that is, they are covered by a white and blurred interpretation of their silhouettes [18]. A next step could be to fully ban and make invisible unwelcome matters. Louis Rosenberg [19], one of AR’s pioneers [20], warns that third parties can not only inject their virtual content into AR but also filter out real-world content that does not meet a reality desired by stakeholders (a government, an international, national, or local company, or a consumer). A patent filed by Apple researchers [21] describes plans to remove avatars (virtual humans) that represent remote people from an AR view if they breach social interaction criteria. These virtual humans and also content in connection with their behavior can be made partially invisible, fully invisible, or anywhere in between along a spectrum of attenuation. We can decide to erase such a person and their activities from our view, for example when they spray virtual graffiti on one of the virtual or real paintings we have on our walls.

5 CONCLUSIONS

The focus of AR research has been on computer vision, computer-generated imagery, and to a lesser extent, modelling of the environment that will be augmented. A multisensory approach is becoming more common [22]. The sensors and actuators of AR devices should be treated as part of the multitude of sensors and actuators offered in ubiquitous computing, which are inseparable from other sensing and processing activities, and which contribute in a similar way to augmented human research. A similar view can be found in some recent papers that discuss how AR relates to ambient intelligence [23], and the Internet of Things [24].

REFERENCES

- [1] R.T. Azuma, 1997. A Survey of Augmented Reality. *Presence: Teleoperators and Virtual Environments* 6, 4, 355–385.
- [2] R.T. Azuma, 2019. The road to ubiquitous consumer augmented reality systems. *Hum Behav & Emerg Tech.* 1:26–32.
- [3] L. Stearns, L. Findlater, and J.E. Froehlich, 2018. Design of an Augmented Reality Magnification Aid for Low Vision Users. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '18)*. Association for Computing Machinery, New York, NY, USA, 28–39. <https://doi.org/10.1145/3234695.3236361>
- [4] U. Gruenefeld, L. Pradel, and W. Heuten, 2019. Locating nearby physical objects in augmented reality. In *Proceedings of the 18th International Conference on Mobile and Ubiquitous Multimedia (MUM '19)*. Association for Computing Machinery, New York, NY, USA, Article 1, 1–10.
- [5] M. Ortega, E. Ivorra, A. Juan, P. Venegas, J. Martínez, and M. Alcañiz, 2021. MANTRA: An Effective System Based on Augmented Reality and Infrared Thermography for Industrial Maintenance. *Appl. Sci.* 2021, 11, 385. <https://doi.org/10.3390/app11010385>
- [6] K. Liliya, H. Pohl, S. Boring, and K. Hornbæk, 2019. Augmented Reality Views for Occluded Interaction. In *ACM Conference on Human Factors in Computing Systems Proceedings (CHI 2019)*, Glasgow, Scotland, UK. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3290605.3300676>
- [7] P. Barnum, Y. Sheikh, A. Datta, and T. Kanade, 2009. Dynamic see-throughs: Synthesizing hidden views of moving objects. *8th IEEE International Symposium on Mixed and Augmented Reality*, Orlando, FL, 111–114. <https://doi.org/10.1109/ISMAR.2009.5336483>
- [8] O. Erat, W. A. Isop, D. Kalkofen, and D. Schmalstieg, 2018. Drone-Augmented Human Vision: Exocentric Control for Drones Exploring Hidden Areas. *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 4, pp. 1437–1446. <https://doi.org/10.1109/TVCG.2018.2794058>
- [9] C. Liu and S. Shen, 2020. An Augmented Reality Interaction Interface for Autonomous Drone. *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 11419–11424. <https://doi.org/10.1109/IROS45743.2020.9341037>
- [10] P. Knierim, T. Kosch, G. LaBorwit, and A. Schmidt, 2020. Altering the Speed of Reality? Exploring Visual Slow-Motion to Amplify Human Perception using Augmented Reality. *Proceedings of the Augmented Humans International Conference (AHs '20)*. Association for Computing Machinery, New York, NY, USA, Article 2, 1–5. <https://doi.org/10.1145/3384657.3384659>
- [11] S. Mori, S. Ikeda, and H. Saito, 2017. A survey of diminished reality: Techniques for visually concealing, eliminating, and seeing through real objects. *IPST Comput Vis Appl* 9, 17. <https://doi.org/10.1186/s41074-017-0028-1>
- [12] Fischer, D. Bartz, and W. Straber, 2005. Stylized augmented reality for improved immersion. *IEEE Proceedings. VR 2005*, 195–202. <https://doi.org/10.1109/VR.2005.1492774>
- [13] B. Bai, Y. Luo, T. Gan, T. *et al.*, 2022. To image, or not to image: class-specific diffractive cameras with all-optical erasure of undesired objects. *eLight* 2, 14. <https://doi.org/10.1186/s43593-022-00021-3>
- [14] B.-K. Seo, J. Park, J.-II. Park, 2019. A New Perspective on Enriching Augmented Reality Experiences: Interacting with the Real World. *Presence*, Vol. 28, 53–63. https://doi.org/10.1162/PRES_a_00341
- [15] R. Sun, H. Haraldsson, Y. Zhao, and S. Belongie, 2019. Anon-Emoji: An Optical See-Through Augmented Reality System for Reducing Appearance Bias in Social Interactions. *CVPR Workshop on Computer Vision for Augmented and Virtual Reality*, Long Beach, CA.
- [16] T. Palladino, 2019. Magic Leap & Weta Workshop Preview Multiplayer Mode for Dr. Grordbort’s Invaders at GDC. 2019. Dr. Grordbort’s Invaders. <https://magic-leap.reality.news/news/magic-leap-weta-workshop-preview-multiplayer-mode-for-dr-grordborts-invaders-gdc-0194957/>
- [17] S. Mann and J. Fung, 2002. EyeTap Devices for Augmented, Deliberately Diminished, or Otherwise Altered Visual Perception of Rigid Planar Patches of Real-World Scenes. *Presence: Teleoperators and Virtual Environments*, 11(2), 158–175. <https://doi.org/10.1162/1054746021470603>
- [18] Koebler, 2014. The ‘Black Mirror’ Technology that Will Soon Be Real. Part 1, December 2014. <https://www.vice.com/en/article/8qxdmz/the-black-mirror-technology-that-will-soon-be-real-part-one>. Part 2, January 2015. <https://www.vice.com/en/article/jp5b3k/digital-slaves-and-memory-erasing-the-black-mirror-technology-being-researched-part-two>
- [19] L.B. Rosenberg, 2022. Augmented Reality: Reflections at Thirty Years. In: Arai, K. (eds) *Proceedings of the Future Technologies Conference (FTC) 2021*, Volume 1. FTC 2021. *Lecture Notes in Networks and Systems*, vol 358. Springer, Cham. https://doi.org/10.1007/978-3-030-89906-6_1
- [20] L.B. Rosenberg, 1992. The use of virtual fixtures as perceptual overlays to enhance operator performance in remote environments. *Interim Report AL/CF-TR-1994-0089*, Armstrong Laboratory.
- [21] A. Bar-Zeev, A. H. Palangie, L.R.D. Centeno, and R. Nair, 2019. Method and device for attenuation of co-user interactions in simulated reality (sr) space. US 2021/0339143 A1. Filed Sept. 17th., 2019. Issued Nov. 4th., 2021.
- [22] K. Karunanayaka, A. Nijholt, T. Halloluw, N. Ranasinghe, M. Wickramasinghe, and D. Vyas, 2021. Multisensory Augmented Reality. In *Human-Computer Interaction – INTERACT 2021: 18th IFIP TC 13 International Conference, Proceedings, Part V*. Springer-Verlag, Berlin, Heidelberg, 558–563. https://doi.org/10.1007/978-3-030-85607-6_77
- [23] R.-D. Vatavu, 2022. Are Ambient Intelligence and Augmented Reality Two Sides of the Same Coin? Implications for Human-Computer Interaction. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3491101>

- [24] A. Arntz, F. Adler, D. Kitzmann, S.C. Eimler (2022). Augmented Reality Supported Real-Time Data Processing Using Internet of Things Sensor Technology. In: Streitz, N.A., Konomi, S. (eds) Distributed, Ambient and Pervasive Interactions.

Smart Living, Learning, Well-being and Health, Art and Creativity. HCII 2022. Lecture Notes in Computer Science, vol 13326. Springer, Cham, 3-17.